

## 1 Summary

Model (Run-name): model1 (test)

Iteration: 2

**TAC: 128009.91260607763**

Took: 0.12374496459960938s

Total Time: 0.5370426177978516s

Epsilons:

- $\varepsilon_{LMTD}$ : 1e-06
- $\varepsilon_{Balancing}$ : 1e-06
- $\varepsilon_{Area}$ : 1e-06
- $\varepsilon_{Beta}$ : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

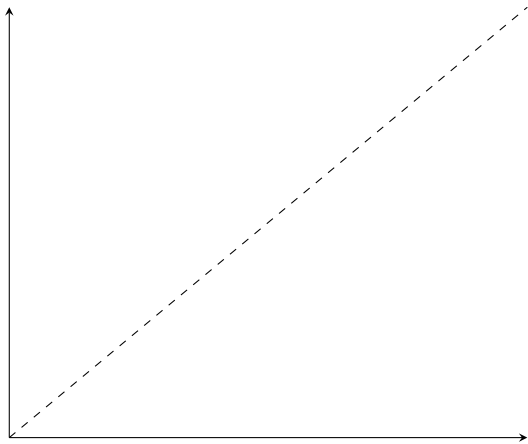
Gamsworld Solution: [http://www.gamsworld.org/minlp/minlplib2/html/heatexch\\_gen1.html](http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html)

## 2 Results

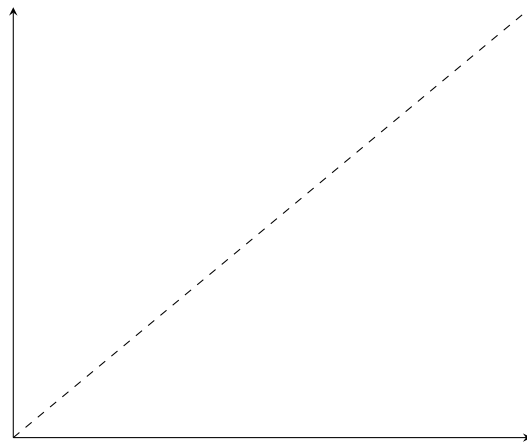
| Variable          | Value              | Variable          | Value                | Variable            | Value              |
|-------------------|--------------------|-------------------|----------------------|---------------------|--------------------|
| $z_{1,1,1}$       | 1.0                | $f_{2,1,2}^H$     | 4.0495788013554295   | $t_{1,1,2}^H$       | 427.7238089674605  |
| $z_{1,1,2}$       | 1.0                | $f_{2,2,2}^H$     | 15.95042119864457    | $t_{2,1,2}^H$       | 370.0              |
| $z_{2,1,2}$       | 1.0                | $f_{1,1,1}^C$     | 8.052417808908414    | $t_{2,2,2}^H$       | 466.68393514135914 |
| $z_{2,2,2}$       | 1.0                | $f_{1,1,2}^C$     | 9.759368610010618    | $t_{1,1,1}^C$       | 557.3655707854455  |
| $z_{cu,1}$        | 1.0                | $f_{2,1,2}^C$     | 5.240631389989382    | $t_{1,1,2}^C$       | 520.6061775801204  |
| $z_{cu,2}$        | 1.0                | $f_{2,2,2}^C$     | 13.0                 | $t_{2,1,2}^C$       | 469.39382241987977 |
| $z_{hu,1}$        | 1.0                | $RecLMTD_{1,1,1}$ | 0.021166404292072658 | $t_{2,2,2}^C$       | 500.0              |
| $A_{1,1,1}^\beta$ | 18.13353447520507  | $RecLMTD_{1,1,2}$ | 0.04321557847449181  | $t_{1,1,1}^{H,in}$  | 1308.5178939476182 |
| $A_{1,1,2}^\beta$ | 13.825772197515043 | $RecLMTD_{2,1,2}$ | 0.06700343198895575  | $t_{1,1,2}^{H,in}$  | 5936.33075337641   |
| $A_{2,1,2}^\beta$ | 9.898970403313276  | $RecLMTD_{2,2,2}$ | 0.00998545173807711  | $t_{2,1,2}^{H,in}$  | 2389.2514927997036 |
| $A_{2,2,2}^\beta$ | 38.943261778500734 | $RecLMTD_{cu,1}$  | 0.010753683011062897 | $t_{2,2,2}^{H,in}$  | 9410.748507200296  |
| $A_{cu,1}^\beta$  | 9.28003023700696   | $RecLMTD_{cu,2}$  | 0.010788088201949427 | $t_{1,1,1}^{H,out}$ | 744.8486473240288  |
| $A_{cu,2}^\beta$  | 21.04660840930274  | $RecLMTD_{hu,1}$  | 0.022585114716274293 | $t_{1,1,2}^{H,out}$ | 4277.238089674605  |
| $A_{hu,1}^\beta$  | 12.595096067947068 | $q_{1,1,1}$       | 563.6692466235894    | $t_{2,1,2}^{H,out}$ | 1498.344156501509  |
| $A_{1,1,1}$       | 18.13353447520507  | $q_{1,1,2}$       | 1659.0926637018051   | $t_{2,2,2}^{H,out}$ | 7460.7485072002955 |
| $A_{1,1,2}$       | 13.825772197515043 | $q_{2,1,2}$       | 890.9073362981949    | $t_{1,1,1}^{C,in}$  | 4670.40232916688   |
| $A_{2,1,2}$       | 9.898970403313276  | $q_{2,2,2}$       | 1950.0               | $t_{1,1,2}^{C,in}$  | 4001.3411301043534 |
| $A_{2,2,2}$       | 38.943261778500734 | $q_{cu,1}$        | 577.2380896746054    | $t_{2,1,2}^{C,in}$  | 2148.6588698956466 |
| $A_{cu,1}$        | 9.28003023700696   | $q_{cu,2}$        | 1559.0926637018051   | $t_{2,2,2}^{C,in}$  | 4550.0             |
| $A_{cu,2}$        | 21.04660840930274  | $q_{hu,1}$        | 486.3307533764106    | $t_{1,1,1}^{C,out}$ | 5234.071575790469  |
| $A_{hu,1}$        | 12.595096067947068 | $t_{1,1}^{(H)}$   | 650.0                | $t_{1,1,2}^{C,out}$ | 5660.433793806159  |
| $dt_{1,1,1}$      | 32.42205022509404  | $t_{1,2}^{(H)}$   | 593.6330753376411    | $t_{2,1,2}^{C,out}$ | 3039.566206193841  |
| $dt_{1,1,2}$      | 13.63307533764106  | $t_{1,3}^{(H)}$   | 427.7238089674605    | $t_{2,2,2}^{C,out}$ | 6500.0             |
| $dt_{1,1,3}$      | 17.723808967460542 | $t_{2,1}^{(H)}$   | 590.0                |                     |                    |
| $dt_{2,1,2}$      | 10.0               | $t_{2,2}^{(H)}$   | 590.0                |                     |                    |
| $dt_{2,1,3}$      | 37.95463318509026  | $t_{2,3}^{(H)}$   | 447.95463318509024   |                     |                    |
| $dt_{2,2,2}$      | 90.0               | $t_{1,1}^{(C)}$   | 617.577949774906     |                     |                    |
| $dt_{2,2,3}$      | 97.95463318509026  | $t_{1,2}^{(C)}$   | 580.0                |                     |                    |
| $dt_{cu,1}$       | 107.72380896746054 | $t_{1,3}^{(C)}$   | 410.0                |                     |                    |
| $dt_{cu,2}$       | 127.95463318509026 | $t_{2,1}^{(C)}$   | 500.0                |                     |                    |
| $dt_{hu,1}$       | 62.42205022509404  | $t_{2,2}^{(C)}$   | 500.0                |                     |                    |
| $f_{1,1,1}^H$     | 2.013104452227105  | $t_{2,3}^{(C)}$   | 350.0                |                     |                    |
| $f_{1,1,2}^H$     | 10.0               | $t_{1,1,1}^H$     | 370.0                |                     |                    |

### 3 *RecLMTD* Tangent Points

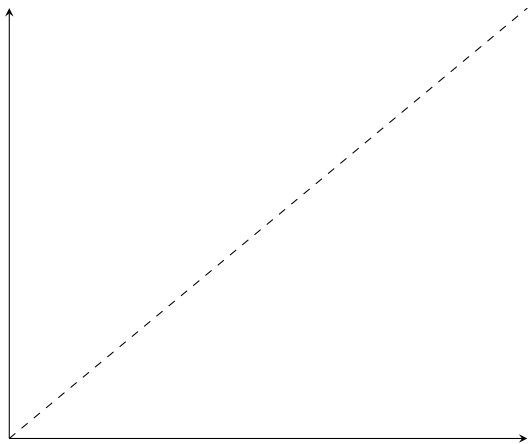
- $(1, 1, 1)$ 
  - Value: 0.021166
  - Error: 0.024942545701250
  - Relative error: 0.540948030802308
- $(1, 1, 2)$ 
  - Value: 0.043216
  - Error: 0.020931845564508
  - Relative error: 0.326308435265960
- $(2, 1, 2)$ 
  - Value: 0.067003
  - Error: 0.019290178835962
  - Relative error: -0.404293934310206
- $(2, 2, 2)$ 
  - Value: 0.009985
  - Error: 0.000661773910576
  - Relative error: 0.062154586782840
- $cu, 1$ 
  - Value: 0.010754
  - Error: 0.000673460667131
  - Relative error: 0.058935171036325
- $cu, 2$ 
  - Value: 0.010788
  - Error: 0.000380283033416
  - Relative error: -0.036538254440593
- $hu, 1$ 
  - Value: 0.022585
  - Error: 0.000014356791480
  - Relative error: 0.000635271115758



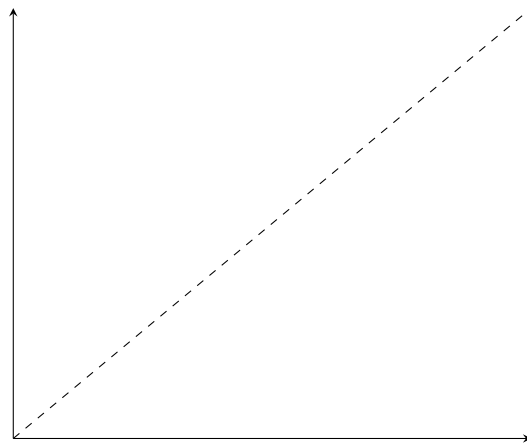
(a) *RecLMTD* - (1, 1, 1)



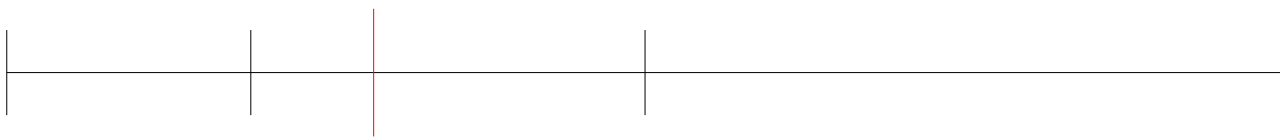
(b) *RecLMTD* - (1, 1, 2)



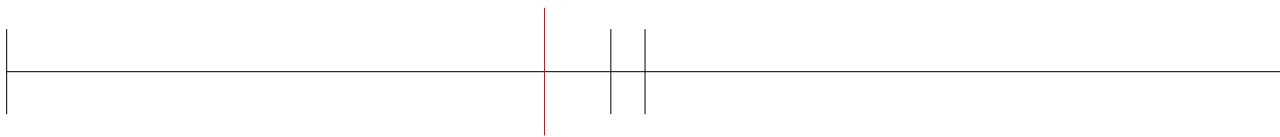
(c) *RecLMTD* - (2, 1, 2)



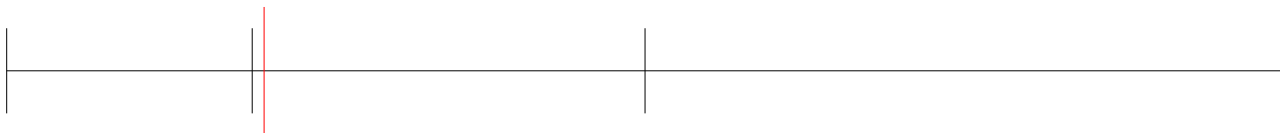
(d) *RecLMTD* - (2, 2, 2)



(a) *RecLMTD* -  $cu, 1$



(b) *RecLMTD* -  $cu, 2$



(a) *RecLMTD* -  $hu, 1$

## 4 Balancing Breakpoints

### 4.1 $b^{H,\text{in}}$

- (1, 1, 1)
  - Value: 1308.5178939476182
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 1, 2)
  - Value: 5936.33075337641
  - Absolute error: 9.094947017729282e-13
  - Relative error: -1.5320822567974912e-16
- (2, 1, 2)
  - Value: 2389.2514927997036
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 2, 2)
  - Value: 9410.748507200296
  - Absolute error: 0.0
  - Relative error: 0.0

### 4.2 $b^{H,\text{out}}$

- (1, 1, 1)
  - Value: 744.8486473240288
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 1, 2)
  - Value: 4277.238089674605
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 1, 2)
  - Value: 1498.344156501509
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 2, 2)
  - Value: 7460.7485072002955
  - Absolute error: 16.9431750546928
  - Relative error: 0.0022761442969934705

**4.3**  $b^{C,in}$ 

- (1, 1, 1)
  - Value: 4670.40232916688
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 1, 2)
  - Value: 4001.3411301043534
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 1, 2)
  - Value: 2148.6588698956466
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 2, 2)
  - Value: 4550.0
  - Absolute error: 0.0
  - Relative error: 0.0

**4.4**  $b^{C,out}$ 

- (1, 1, 1)
  - Value: 5234.071575790469
  - Absolute error: 745.9311275253449
  - Relative error: 0.16620048684387365
- (1, 1, 2)
  - Value: 5660.433793806159
  - Absolute error: 579.6462061531183
  - Relative error: 0.11408589636018877
- (2, 1, 2)
  - Value: 3039.566206193841
  - Absolute error: 579.6462061531174
  - Relative error: 0.23563620204865257
- (2, 2, 2)
  - Value: 6500.0
  - Absolute error: 0.0
  - Relative error: 0.0



(a)  $t^{(H)} - (1, 1)$



(a)  $t^{(H)} - (1, 2)$



(a)  $t^{(H)} - (2, 2)$



(a)  $t^{(H)} - (2, 2)$



(a)  $t^{(H)} - (1, 1, 1)$



(a)  $t^{(H)} - (1, 1, 2)$



(a)  $t^{(H)} - (2, 1, 2)$



(a)  $t^{(H)} - (2, 2, 2)$



(a)  $t^{(C)} - (1, 2)$



(a)  $t^{(C)} - (1, 3)$



(a)  $t^{(C)} - (1, 3)$



(a)  $t^{(C)} - (2, 3)$



(a)  $t^{(C)} - (1, 1, 1)$



(a)  $t^{(C)} - (1, 1, 2)$



(a)  $t^{(C)} - (2, 1, 2)$



(a)  $t^{(C)} - (2, 2, 2)$

## 5 Area Breakpoints

### 5.1 $A$

- $(1, 1, 1)$ 
  - Value: 18.133534
  - Error: 5.728167846880737
  - Relative error: 0.240056965323001
- $(1, 1, 2)$ 
  - Value: 13.825772
  - Error: 129.571526211802961
  - Relative error: 0.903584151508557
- $(2, 1, 2)$ 
  - Value: 9.898970
  - Error: 109.488727828922393
  - Relative error: 0.917085507553236
- $(2, 2, 2)$ 
  - Value: 38.943262
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000

### 5.2 $A_{CU}$

- $cu, 1$ 
  - Value: 9.280030
  - Error: 3.134840639537449
  - Relative error: 0.252506906492290
- $cu, 2$ 
  - Value: 21.046608
  - Error: 12.592649932751964
  - Relative error: 0.374343863491456

### 5.3 $A_{HU}$

- $hu, 1$ 
  - Value: 12.595096
  - Error: 0.585506958122934
  - Relative error: 0.044421864232225



(a)  $A^\beta - (1, 1, 1)$



(a)  $A^\beta - (1, 1, 2)$



(a)  $A^\beta - (2, 1, 2)$



(a)  $A^\beta - (2, 2, 2)$



(a)  $A^\beta - cu, 1$



(a)  $A^\beta - cu, 2$

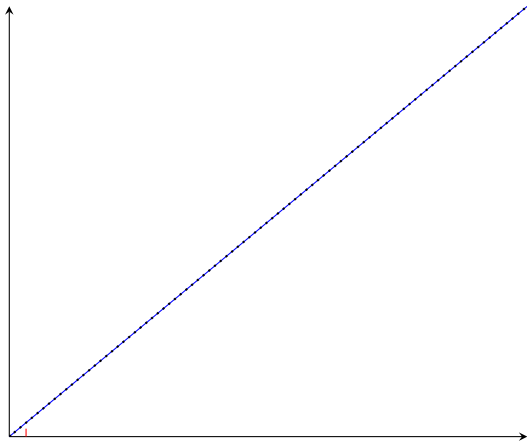


(a)  $A^\beta - hu, 1$

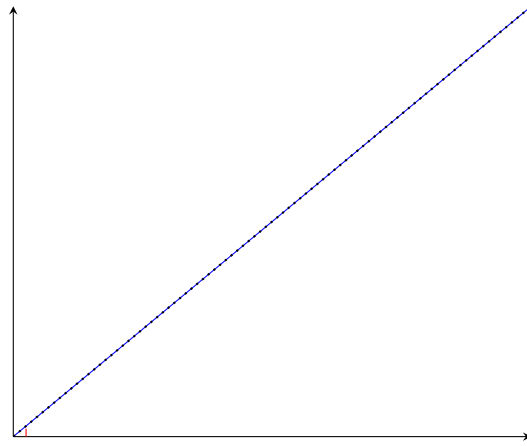
## 6 $A^\beta$ Breakpoints

- (1,1,1)
  - $q$ : 18.133534
  - $\hat{q}^\beta$ : 18.133534
  - $q^\beta$ : 18.133534
  - Breakpoints: [0, 5.0, 560.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (1,1,2)
  - $q$ : 13.825772
  - $\hat{q}^\beta$ : 13.825772
  - $q^\beta$ : 13.825772
  - Breakpoints: [0, 560.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (2,1,2)
  - $q$ : 9.898970
  - $\hat{q}^\beta$ : 9.898970
  - $q^\beta$ : 9.898970
  - Breakpoints: [0, 28.33333333333332, 720.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (2,2,2)
  - $q$ : 38.943262
  - $\hat{q}^\beta$ : 38.943262
  - $q^\beta$ : 38.943262
  - Breakpoints: [0, 390.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- $cu, 1$ 
  - $q$ : 9.280030
  - $\hat{q}^\beta$ : 9.280030
  - $q^\beta$ : 9.280030
  - Breakpoints: [0, 2.9819181007907165, 181.61828057849593]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- $cu, 2$

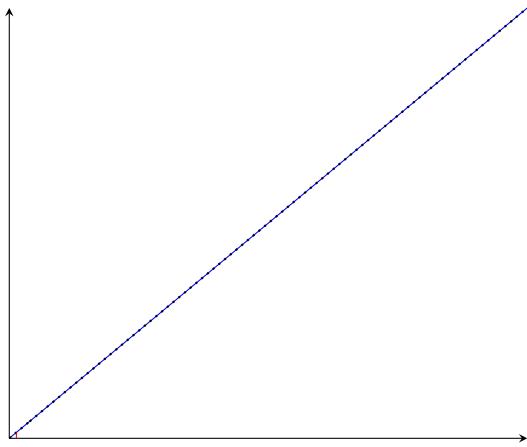
- $q$ : 21.046608
- $\hat{q}^\beta$ : 21.046608
- $q^\beta$ : 21.046608
- Breakpoints: [0, 24.97364426355679, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$ 
  - $q$ : 12.595096
  - $\hat{q}^\beta$ : 12.595096
  - $q^\beta$ : 12.595096
  - Breakpoints: [0, 4.943755299006499, 237.3002543523117]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000



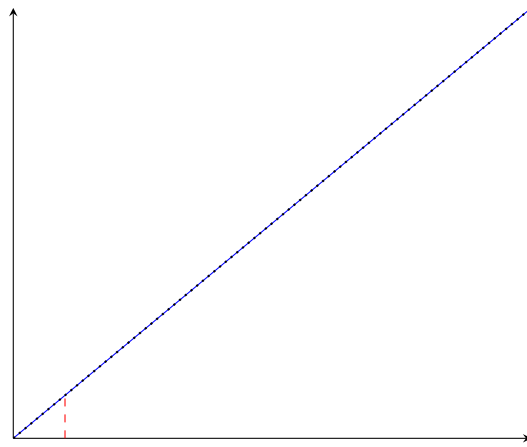
(a)  $q^\beta - (1, 1, 1)$



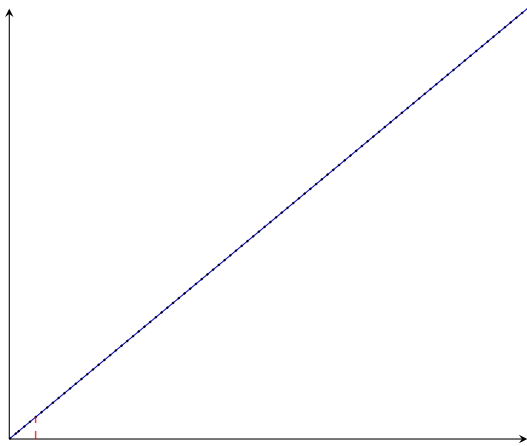
(b)  $q^\beta - (1, 1, 2)$



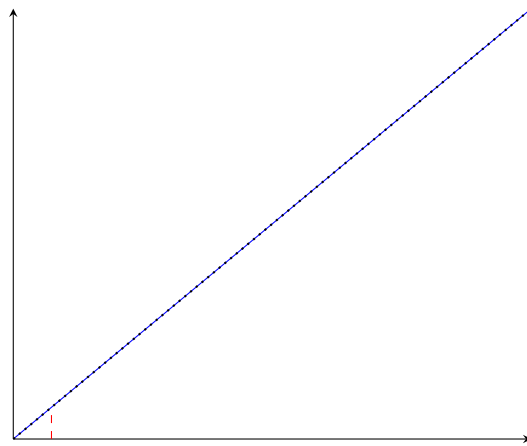
(c)  $q^\beta - (2, 1, 2)$



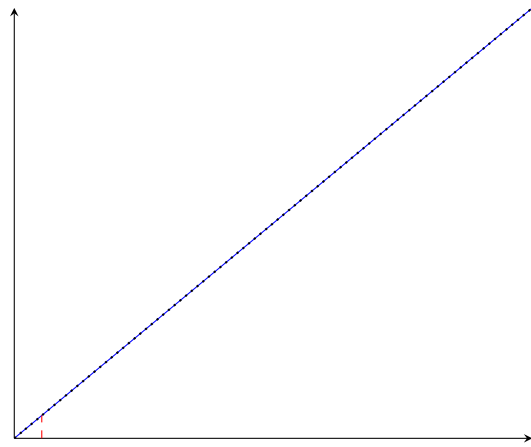
(d)  $q^\beta - (2, 2, 2)$



(a)  $q^\beta - cu, 1$



(b)  $q^\beta - cu, 2$



## 7 Other variables

| Variable          | Value                | Variable          | Value                |
|-------------------|----------------------|-------------------|----------------------|
| $z_{1,2,1}$       | 0.0                  | $RecLMTD_{hu,2}$  | 0.005549329826813255 |
| $z_{1,2,2}$       | -0.0                 | $q_{1,2,1}$       | 0.0                  |
| $z_{2,1,1}$       | -0.0                 | $q_{1,2,2}$       | 0.0                  |
| $z_{2,2,1}$       | -0.0                 | $q_{2,1,1}$       | 0.0                  |
| $z_{hu,2}$        | -0.0                 | $q_{2,2,1}$       | 0.0                  |
| $A_{1,2,1}^\beta$ | 0.0                  | $q_{hu,2}$        | 0.0                  |
| $A_{1,2,2}^\beta$ | 0.0                  | $H_{1,2,1}$       | 650.0                |
| $A_{2,1,1}^\beta$ | 0.0                  | $H_{1,2,2}$       | 370.0                |
| $A_{2,2,1}^\beta$ | 0.0                  | $H_{2,1,1}$       | 590.0                |
| $A_{hu,2}^\beta$  | 0.0                  | $H_{2,2,1}$       | 590.0                |
| $A_{1,2,1}$       | 0.0                  | $C_{1,2,1}$       | 500.0                |
| $A_{1,2,2}$       | 0.0                  | $C_{1,2,2}$       | 350.0                |
| $A_{2,1,1}$       | 0.0                  | $C_{2,1,1}$       | 580.0                |
| $A_{2,2,1}$       | 0.0                  | $C_{2,2,1}$       | 350.0                |
| $A_{hu,2}$        | 0.0                  | $H_{1,2,1}^{in}$  | 5191.482106052382    |
| $dt_{1,2,1}$      | 150.0                | $H_{1,2,2}^{in}$  | 0.0                  |
| $dt_{1,2,2}$      | 223.63307533764112   | $H_{2,1,1}^{in}$  | 11800.0              |
| $dt_{1,2,3}$      | 207.72380896746054   | $H_{2,2,1}^{in}$  | 0.0                  |
| $dt_{2,1,1}$      | 24.166666666666668   | $H_{1,2,1}^{out}$ | 5191.482106052382    |
| $dt_{2,2,1}$      | 90.0                 | $H_{1,2,2}^{out}$ | 0.0                  |
| $dt_{hu,2}$       | 180.0                | $H_{2,1,1}^{out}$ | 11800.0              |
| $f_{1,2,1}^H$     | 7.986895547772895    | $H_{2,2,1}^{out}$ | 0.0                  |
| $f_{1,2,2}^H$     | 0.0                  | $C_{1,2,1}^{in}$  | 6500.0               |
| $f_{2,1,1}^H$     | 20.0                 | $C_{1,2,2}^{in}$  | 0.0                  |
| $f_{2,2,1}^H$     | 0.0                  | $C_{2,1,1}^{in}$  | 4029.5976708331195   |
| $f_{1,2,1}^C$     | 13.0                 | $C_{2,2,1}^{in}$  | 0.0                  |
| $f_{1,2,2}^C$     | 0.0                  | $C_{1,2,1}^{out}$ | 6500.0               |
| $f_{2,1,1}^C$     | 6.947582191091586    | $C_{1,2,2}^{out}$ | 0.0                  |
| $f_{2,2,1}^C$     | 0.0                  | $C_{2,1,1}^{out}$ | 4029.5976708331195   |
| $RecLMTD_{1,2,1}$ | 0.005127303322837854 | $C_{2,2,1}^{out}$ | 0.0                  |
| $RecLMTD_{1,2,2}$ | 0.004270239531638324 |                   |                      |
| $RecLMTD_{2,1,1}$ | 0.029166666666666688 |                   |                      |
| $RecLMTD_{2,2,1}$ | 0.010239999999999999 |                   |                      |